Do US Macro Releases Move Bitcoin?

A high-frequency causal analysis of CPI, NFP and FOMC announcements

Alexandro Bizeul · Julius Hayter · Léonard Schwob

MSc 2, Albert School × Mines Paris · Causal Inference & Experimentation (MSA-DAT09-04) · November 2026

Abstract. We estimate the causal effect of scheduled US macroeconomic releases (CPI and NFP at 08:30 New York time, FOMC decisions at 14:00) on Bitcoin in a ±120-minute window, using 2,517,120 one-minute BTCUSDT bars from January 2022 to October 2026 and 151 releases (56 CPI, 57 NFP, 38 FOMC). Release dates are fixed in advance by the agencies, which makes their timing plausibly exogenous to Bitcoin. Two effects are identified. (i) **Volatility**: realised volatility in the 30 minutes after a release is ×2.4 higher than on comparable days without a release (t = 17); the effect vanishes when the clock is shifted. (ii) **Direction**: the BTC reaction follows the surprise (actual minus consensus), with a slope of −65 bps per standard deviation of surprise at +5 minutes and −80 at +30 minutes, negative in all 20 sub-samples we examine. The parallel-trends assumption of the difference-in-differences (DiD) is **rejected**: Bitcoin drifts up by about 23 bps in the two hours before the release. We therefore use a synthetic control (a "synthetic day" built from non-release days) as the main estimator and keep the DiD as a comparison and falsification device. The sign of the directional effect is robust, its magnitude is not (it is three times larger in 2022 than afterwards), and the cause of the pre-release drift is not demonstrated.

1. Introduction and causal question

Bitcoin trades around the clock, yet a large share of its short-run variance seems to cluster around US macro announcements. The question is whether this clustering is *caused* by the releases or simply reflects the fact that 08:30 New York is an active hour and that risk-on or risk-off regimes coincide with the macro calendar. A naive comparison of BTC around release time with BTC at other times mixes the two. The fundamental problem of causal inference applies directly: on a release day we observe Bitcoin *with* the release, never *without* it, so the counterfactual path must be built.

We ask two questions. **Q1 (risk):** do releases change how much Bitcoin moves? **Q2 (direction):** does the sign and size of the surprise change where Bitcoin goes? The first needs no consensus data and is the more robustly identified; the second needs a surprise measure and is more fragile. The estimand is the effect on release days (an ATT, Section 3). The practical decision behind the study is the one a trading or risk desk with crypto exposure faces: should it treat these releases as a risk trigger on crypto exposure, and can it extract a directional signal from them? (Section 8.)

2. Data

Prices. Binance public archive (data.binance.vision), BTCUSDT spot, one-minute klines from 2022-01-01 to 2026-10-15 (2,517,120 bars). Gaps (0.003% of minutes) are forward-filled for at most three minutes. For each event we build a path on a 5-minute grid from −120 to +120 minutes: cumulative return in basis points (bps) relative to the price at the release time t0 (so the path equals 0 at t0). Realised volatility RV over a 30-minute window is the square root of the sum of the six squared 5-minute returns on that grid.

Calendar. CPI and NFP release dates come from the FRED release calendar; FOMC decision dates from the Federal Reserve calendar (38 meetings). Five FRED dates that correspond to unscheduled annual revisions rather than actual releases were removed; they are listed in the notebook, and checking them against the BLS calendar is still pending. The late-2025 releases displaced by the government shutdown are kept at their true dates. Releases at 14:00 other than FOMC decisions (FOMC minutes) are excluded from the donor pool.

Surprises. Without a Bloomberg terminal, we reconstructed them from public sources. NFP: Investing.com consensus (57 of 57 releases). CPI: Investing.com consensus for *core* CPI m/m (55 of 56; the 2025-12-18 release has no consensus), so the "CPI" surprise is the core surprise; 35% of CPI surprises are exactly zero because of rounding to 0.1 pt. FOMC: the Bauer–Swanson monetary-policy surprise series, which ends in December 2023 and therefore covers 16 of 38 decisions. Surprises are standardised within each event type (mean 0, std. dev. 1), so a "standard deviation" of CPI and of NFP are not the same quantity; we compare signs across types, not levels. In total 128 of 151 events have a surprise.

Source	Content	Coverage
Binance	BTCUSDT 1-min klines	2,517,120 bars, 2022-01 to 2026-10
FRED / Federal Reserve	CPI, NFP and FOMC dates	56 CPI, 57 NFP, 38 FOMC
Investing.com (manual copy)	Consensus and actual: NFP, core CPI m/m	57/57 NFP, 55/56 CPI
Bauer–Swanson (FRBSF)	FOMC policy surprise	16/38 decisions (2022–2023)

3. Causal framework

Unit and treatment. The unit is an event window e: a release at time t0 (08:30 New York for CPI and NFP, 14:00 for FOMC). The treatment $D_e = 1$ means a release occurs at t0. The potential outcomes are $Y_e(\tau; 1)$, the BTC path in bps relative to t0 when there is a release, and $Y_e(\tau; 0)$, the path on the same day had there been none. Only one is observed.

Estimands. (1) The volatility effect, $\mathrm{ATT}_{\mathrm{vol}} = \mathrm{E}[\ln \mathrm{RV}_{[0,30]} - \ln \mathrm{RV}_{[-30,0]} \mid D=1] - \mathrm{E}[\cdot \mid D=0]$, expressed as a ratio $\exp(\mathrm{ATT}_{\mathrm{vol}})$. (2) The directional effect, the slope β of the estimated counterfactual gap $\hat{\tau}_e(h) = Y_e(h; 1) - \hat{Y}_e(h; 0)$ on the standardised surprise z_e, in bps per standard deviation.

ATE, ATT, ATU. What we estimate is the **ATT**: the effect of a release on the days on which a release actually occurs. The **ATE** and the **ATU** (the effect a release would have had on a day without one) are not identified here. Release dates are set by the agencies, so no day without a release carries one, and what a release would do on such a day (a quiet Tuesday, with a different set of numbers) would require extrapolating beyond the support of the data. Our donor days are used only as the source of the counterfactual for treated days, and we state this explicitly rather than quote an ATE.

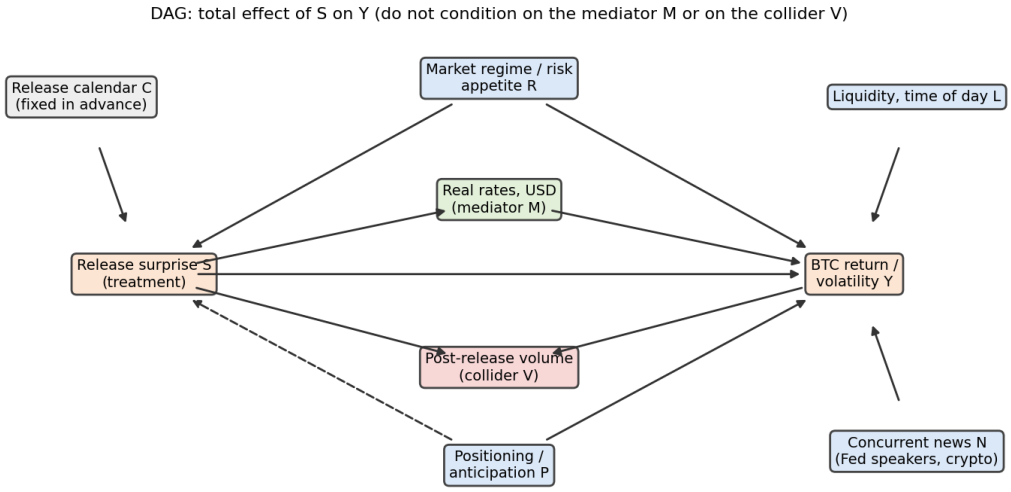


Figure 1. DAG. The surprise S affects Bitcoin Y directly and through real rates and the dollar (mediator M); we want the total effect, so M is not controlled. The market regime R is a confounder, closed by using local donors (±180 days) and pre-release covariates. Post-release volume V is a collider and is never conditioned on. Positioning P and concurrent news N are not observed and are the main threats. The calendar C is exogenous.

Backdoor criterion. The backdoor paths from S to Y are S ← R → Y (regime) and S ← P → Y (positioning). The first is closed by comparing each release with donor days from the same regime (±180 days) and, for the synthetic day, by matching the pre-release path and covariates (24-hour volatility, 24-hour return, volume). The second is not closed by design: it is the object of the pre-trend and anticipation tests below. Identification rests on four assumptions: (i) the calendar is exogenous; (ii) no anticipation beyond what the pre-window captures; (iii) parallel trends, or a good pre-release fit for the synthetic day; (iv) no concurrent shock specific to release days.

4. Estimation strategy

Donor pool. For each event, donors are business days within ±180 days that have no release at the same clock time, excluding US holidays, Thursdays at 08:30 (jobless claims) and days with other 08:30 releases; the median event has 68 donors. Placebo clocks (a release at 08:30 on a donor day) therefore carry no announcement.

DiD with equal weights. For each event, the gap is the treated path minus the average donor path, both relative to t0. The event study averages these gaps over events; the pre-release gap should be zero if parallel trends hold. We test this jointly with the maximum absolute t-statistic over the 24 pre-release points (sign-flip permutation), and calibrate the reference distribution by *placebo in space*: we draw as many random donor days as there are events, treat them as fake releases with the same ±180-day window (hence the same donor sharing), and recompute the statistic. A Wald test is also reported but is fragile because the 24 points are strongly correlated.

Synthetic day (main method). For each event we build a counterfactual path as a weighted combination of donor-day paths. Weights come from a ridge regression on the pre-release window (normalised by its volatility) plus covariates, with the penalty λ selected on placebo days only, never on treated days. Because the donor pool has more days than the window has points, the RMSPE ratio is mechanically large; we therefore rely only on the comparison with **in-space placebos**, where every donor day in turn plays the fake treated unit and the same procedure is applied. As a robustness check we also run the classical synthetic control with non-negative weights summing to one.

Directional effect and inference. At horizon h (5, 15, 30, 60 minutes) we regress the counterfactual gap on the standardised surprise. The p-value is a **Fisher permutation test** (3,000 shuffles of the surprise within each event type); we also report heteroskedasticity-robust (HC1) standard errors, which do not correct for donor sharing and are therefore secondary. Because four horizons are tested, we apply a Bonferroni factor of 4.

Falsification and power. We repeat the volatility analysis with the clock shifted by −12 h, −6 h and +6 h (no release occurs then); the effect should vanish. We also compute the minimal detectable effect (80% power, 5% level).

5. Descriptive evidence

Figure 2a shows the average absolute BTC path: it is flat before t0 and rises sharply afterwards on release days, much more than on donor days. Figure 2b shows that the volatility effect is not driven by a few extreme events: the distribution of the event-level effect lies almost entirely to the right of zero for each type. The first column of Table 1 gives the sample sizes.

Naive comparisons versus DiD. Table 0 is the 2×2 table of the course's tutoring example, applied to the log of realised volatility in the 30 minutes before and after the release time, release days versus all donor days. The naive “after only” comparison mixes the effect with a selection bias: FOMC days are already more volatile than donor days *before* 14:00 (difference 0.28 in logs), so it overstates the effect (0.89 versus a DiD of 0.62). The naive before/after comparison on release days only would be biased if volatility rose at that hour anyway; on donor days it does not (−0.004 and −0.005), so here it is close to the DiD. The matched DiD of Section 6 uses, for each release, only donors within ±180 days.

Clock	Releases / donors	Release days: before → after	Donor days: before → after	Naive after only	Naive before vs after	DiD 2×2
08:30	113 / 326	3.38 → 4.34	3.30 → 3.29	1.05	0.97	0.97
14:00	38 / 1120	3.70 → 4.31	3.42 → 3.42	0.89	0.61	0.62

Table 0. Mean log realised volatility (30 minutes) before and after the release clock. Differences are in logs: exp(0.97) = ×2.6, exp(0.62) = ×1.9.

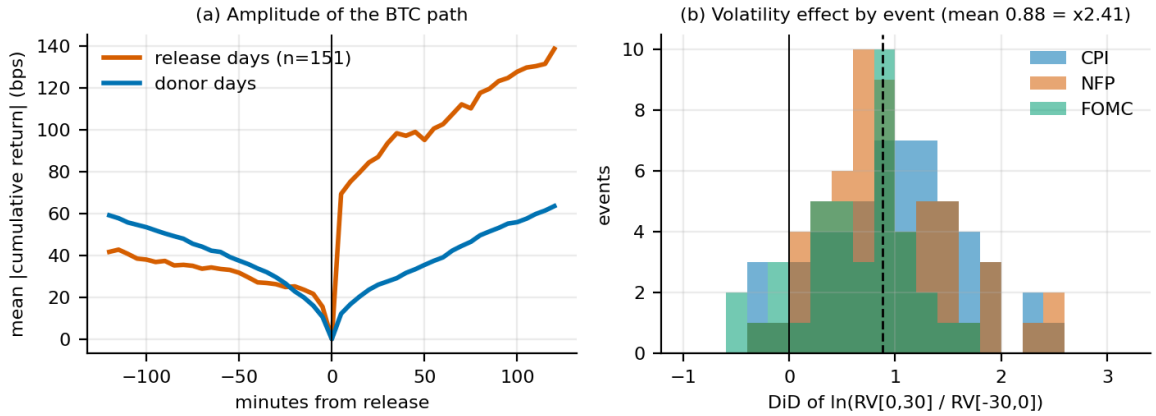


Figure 2. (a) Mean absolute cumulative return around t0, release days versus donor days. (b) Event-level DiD of the log change in realised volatility (30 minutes after versus 30 minutes before), by event type; the dashed line is the mean.

6. Results

6.1 Volatility: a large, falsifiable effect

The DiD volatility effect is 0.88 (SE 0.05, t = 17.2), i.e. realised volatility in the 30 minutes after a release is **×2.41** that of comparable days (95% CI ×2.18 to ×2.66). With the synthetic day, 41% of events are significant at 5% against the 5% expected by chance (44% with the classical version). The ridge and classical versions agree: the mean absolute gap in the first 30 minutes exceeds the placebo benchmark by a factor 2.5 and 2.4, and the event-level gaps correlate at 0.99. The classical version fits the pre-window slightly worse (median RMSPE 0.65 vs 0.46) and relies on very few donors (about 5 effective donors versus 44), so its agreement is reassuring rather than additional independent evidence.

Group	Events	Volatility ratio	Significant at 5% (synthetic day)
All events	151	×2.41	41%
CPI	56	×2.74	48%
NFP	57	×2.53	37%
FOMC	38	×1.85	37%
Before spot ETFs (< 2024-01-11)	65	×2.50	51%
After spot ETFs	86	×2.34	34%

Table 1. Volatility effect by group. The ETF split is exploratory (not pre-registered).

Falsification. With the clock shifted by −12, −6 and +6 hours, the DiD volatility effect is 0.04 (t = 0.7), 0.03 (t = 0.5) and -0.03 (t = -0.6), against 0.97 (t = 16.1) at the true clock (the 113 CPI/NFP events at 08:30). The effect is tied to the release, not to an artefact of the calendar of days or of the computation (Figure 4c).

6.2 Parallel trends are rejected

The DiD event study (Figure 3a) shows a clear pre-release gap: relative to t0, Bitcoin on release days is 23 bps higher two hours before the release than on donor days, and 20 bps higher one hour before. The joint test gives max|t| = 5.64; under the placebo null its median is 2.1 and its 95th percentile 3.67, so the calibrated p-value is 0.002 (Figure 3b). The same test run separately by clock, each against its own placebo pool, gives −16 bps at −60 minutes and p = 0.003 for the 113 releases at 08:30, and −30 bps and p = 0.010 for the 38 FOMC decisions at 14:00. The naive standard errors are 1.2 to 1.6 times too small because neighbouring events share donors. **The equal-weight DiD is therefore not credible for signed effects.** The signed ATT is nonetheless reported for completeness; every confidence interval contains zero:

Horizon (min)	ATT (bps)	95% CI
5	−4.4	[−21.4, 12.6]
15	−14.4	[−34.1, 5.4]
30	−9.3	[−33.0, 14.4]
60	−0.4	[−25.5, 24.7]
120	−0.4	[−31.1, 30.3]

Table 2. Signed DiD ATT averaged over all releases. Bad and good surprises offset each other, and the pre-trend violation applies.

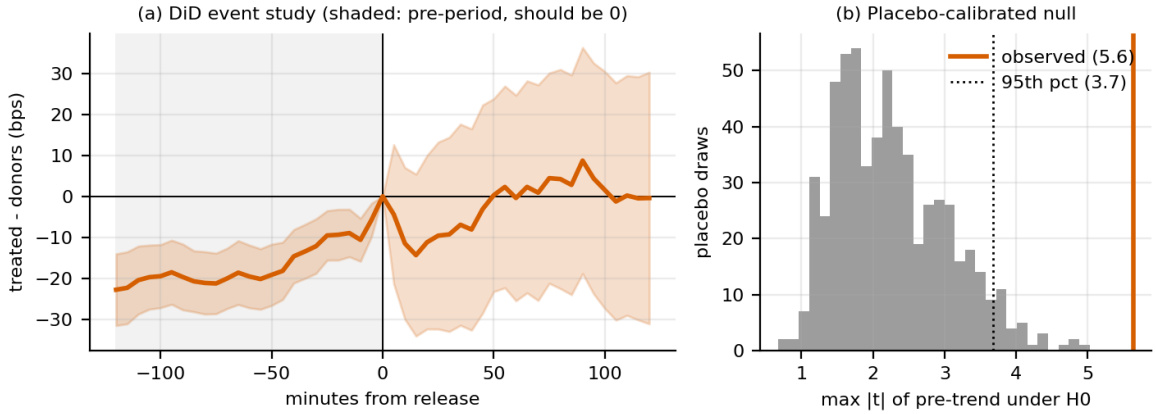


Figure 3. (a) DiD event study with 95% confidence band. The grey zone is the pre-period, where the gap should be zero under parallel trends. (b) Null distribution of the pre-trend statistic from in-space placebos and the observed value.

6.3 The pre-release drift

The drift is not spread over the night. Hour by hour (Figure 4a, b), it is concentrated in the **last hour before the release**: 16.2 bps of excess return (placebo-calibrated t = 3.0). Twenty-six hours are tested, so one other significant hour (the evening before) is likely noise. The drift survives changes of the donor definition: the gap at −60 minutes is −23 to −16 bps for pools of ±30 to ±365 days and for the same weekday only (all max|t| above 4). It is specific to the true release clock: with shifted clocks no comparable pre-trend appears after calibration. It is stable in sign across years and larger before CPI than before NFP. We read it as positioning or information circulating before the release, but this is an interpretation, not a demonstrated cause.

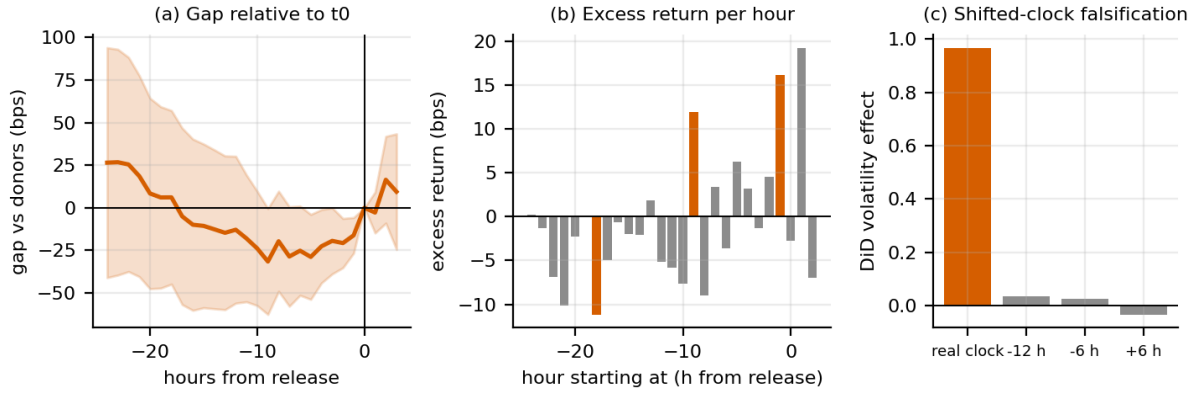


Figure 4. (a, b) The 113 CPI/NFP releases at 08:30. (a) Gap versus donors relative to t0 by hour, with placebo-calibrated 95% band. (b) Excess return per hour (orange where |t| ≥ 2). (c) Falsification: DiD volatility effect at the true clock and at shifted clocks (113 CPI/NFP events at 08:30).

6.4 Synthetic day: counterfactual and inference

Figure 5 illustrates the method on one event. The synthetic day reproduces the pre-release path closely, and the post-release gap of the actual event lies outside the cloud of 40 in-space placebo gaps drawn from donor days. Given the diagnosis in 6.2, signed effects below rely on this estimator, not on the equal-weight DiD.

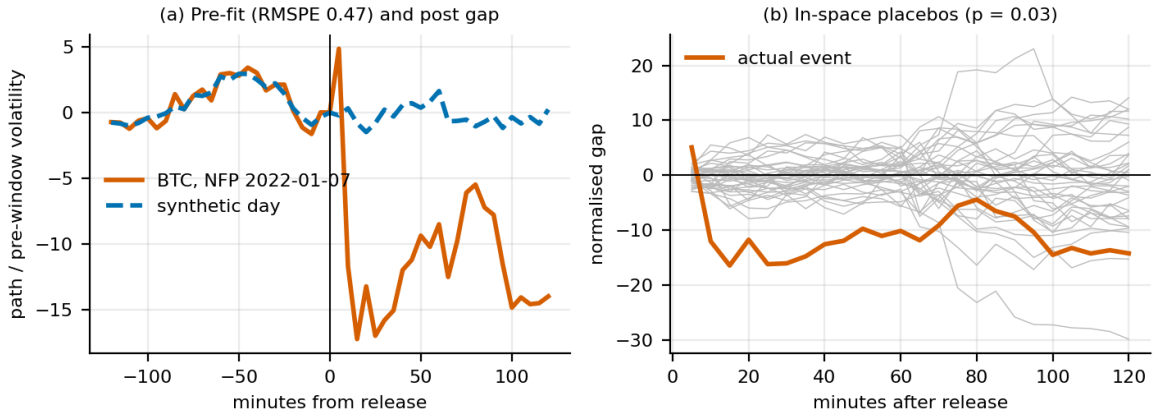


Figure 5. Synthetic-day construction for one event. (a) Actual path and synthetic day (both divided by the pre-window volatility). (b) Post-release gap of the actual event against in-space placebos.

6.5 Direction: the reaction follows the surprise

With 128 surprises, the slope of the gap on the standardised surprise is **−65 bps per standard deviation at +5 minutes** (95% CI −87 to −42, permutation p < 0.001) and −80 at +30 minutes (95% CI −117 to −43). The equal-weight DiD gives almost the same figures (−65 and −79), so the pre-trend does not drive the slope: a constant pre-release drift does not correlate with the later surprise. A higher-than-expected number (stronger payrolls, higher core inflation, a more hawkish Fed) lowers Bitcoin, the usual "good news is bad news" reading for a risk asset sensitive to rates.

Sub-sample	n	Slope +5 min	p (perm.)	Slope +30 min	p (perm., ×4)
All types	128	−65	0.001	−80	0.001
CPI (core)	55	−93	0.001	−123	0.001
NFP	57	−26	0.004	−29	0.087
FOMC (2022-23)	16	−110	0.004	−116	0.004
Excluding 2022	96	−32	0.001	−28	0.059
2022 only	32	−98	0.001	−128	0.003
Before spot ETFs	65	−78	0.001	−98	0.001
After spot ETFs	63	−42	0.001	−40	0.017
Excluding 5 largest surprises	123	−73	0.001	−90	0.001

Table 3. Directional slope (bps per standard deviation of surprise) by sub-sample, synthetic day. p-values are Fisher permutation p-values multiplied by 4 (four horizons tested), capped at 1.

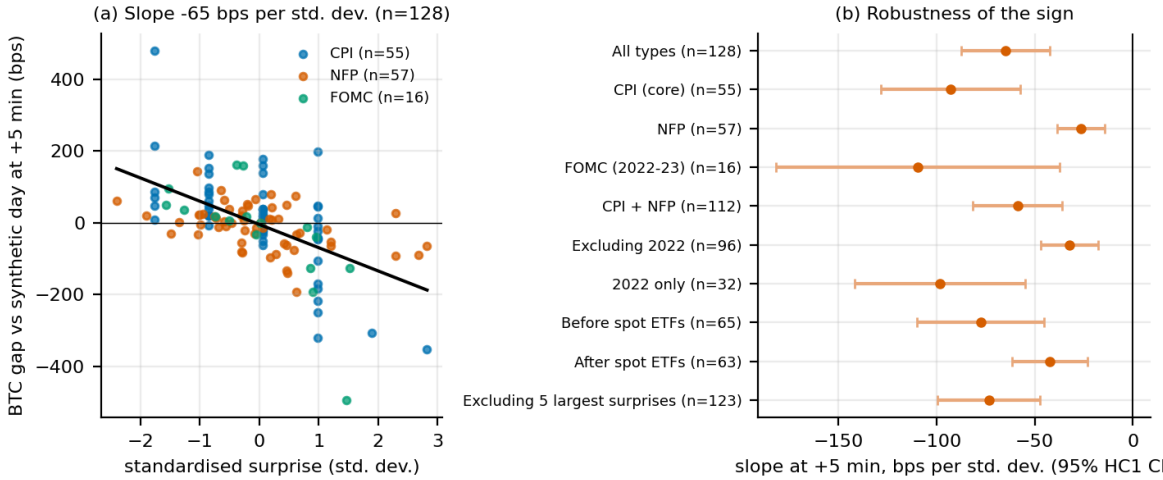


Figure 6. (a) Surprise versus BTC gap at +5 minutes, by event type; the line is the pooled OLS slope. (b) Slope at +5 minutes by sub-sample with 95% HC1 intervals; the sign is negative in every row.

The sign is negative in all 20 sub-sample/horizon combinations, and 18 of 20 remain significant at 5% after the Bonferroni factor; the exceptions are NFP and the post-2022 sub-sample at 30 minutes (adjusted p between 0.06 and 0.09). With the 128 releases that have a surprise, the minimal detectable effect is 28 bps per std. dev. at 5 minutes and 39 at 30 minutes; the pooled slopes are about twice that threshold. **Outliers.** Some 2022 reactions exceed ±500 bps. Estimators that limit

their weight give smaller but clearly negative slopes at +5 minutes: −49 with the reaction winsorised at 5%, −49 with Theil–Sen, and a Spearman rank correlation of −0.58 ($p < 0.001$). **Placebo in time.** The surprise is unknown before t0, so it should not explain the path before the release. The slope of the pre-release gap on the surprise is −8.5 bps/σ at −60 minutes (permutation $p = 0.04$) and −5.8 at −30 minutes ($p = 0.13$), about eight times smaller than the post-release slope. Its sign is the opposite of what an information leak would produce (Bitcoin rises slightly *more* before hawkish surprises), so we read it as weak noise, but we report it. Two caveats matter. First, the pooled slope is pulled by CPI and FOMC, whose standardised surprises are not comparable with NFP's; the per-type slopes are the informative ones. Second, the effect is not stable: it is −98 bps/σ in 2022 and −32 afterwards.

7. Identification critique and limits

Assumption	Verdict	Evidence and caveat
Calendar exogenous	Plausible for the date	Dates set in advance. The content (the surprise) depends on the economy, hence on the regime; we absorb it with local donors and sub-period checks.
Parallel trends (DiD)	Rejected	max|t| = 5.6, calibrated p = 0.002; robust to donor definition. DiD kept for volatility and as comparison.
No anticipation	Violated in appearance	Drift concentrated in the last hour, specific to the true clock. Cause not demonstrated (positioning, leaked information or calendar effect).
No interference (SUTVA)	Partly violated	Neighbouring events share donors; calibrated SEs are 1.2–1.6 times the naive ones. Permutation p-values are the reference; HC1 SEs ignore sharing.
No concurrent news	Not verifiable	Fed speakers, crypto news and ETF flows are unobserved; bias possible in both directions.
Stable regime	Probably not	Volatility ratio ×2.50 before and ×2.34 after spot ETFs; directional slope three times larger in 2022. Exploratory.
Synthetic-day fit	Acceptable	Median pre RMSPE 0.46 (ridge) and 0.65 (classical); both give 41–44% significant events.
Surprise measurement	Imperfect	Public consensus, core CPI only, 35% zero CPI surprises, FOMC surprise for 16 of 38 decisions. Measurement error probably attenuates slopes.
External validity	Limited	Two-hour window, one venue (Binance BTCUSDT), one asset; no long-run effect studied.

What the analysis does not do. It does not identify the cause of the pre-release drift, an effect beyond about an hour, or an effect on assets other than Bitcoin. The surprise slopes are partly exploratory: four horizons and many sub-samples were examined, which is why we report Bonferroni-adjusted p-values and the sign across all sub-samples rather than a single headline estimate. A month-block bootstrap for standard errors and the “honest DiD” bounds of Rambachan and Roth (which would bound the effect despite the violated pre-trend) are natural extensions that we have not run.

8. Conclusion and recommendation

Findings. (1) A release multiplies Bitcoin's 30-minute realised volatility by about 2.4 (CPI 2.7, NFP 2.5, FOMC 1.9); the effect disappears with shifted clocks and holds under two synthetic-control variants. (2) The price reacts to the surprise with a negative slope (−65 bps/σ at 5 minutes, −80 at 30), stable in sign but not in size. (3) Parallel trends fail because Bitcoin drifts up by about 23 bps in the two hours before the release; this is why the main estimator is the synthetic day.

Recommendation (one page version). *Treat CPI, NFP and FOMC as risk events on crypto exposure; do not yet treat them as a systematic directional signal.* Concretely: reduce size or leverage on Bitcoin before 08:30 New York (and 14:00 on FOMC days), widen stops and risk limits for the first 30 minutes, and avoid executing large orders in that window. On direction, a number above consensus argues for hedging a long position rather than opening a short, but a systematic strategy would need out-of-sample validation on the next releases, execution costs, and a control asset (ETH, gold) to see whether the pre-release drift is specific to Bitcoin.

Practical relevance beyond Bitcoin. The design carries over to any market that reacts to scheduled announcements: timing is set in advance, non-event days at the same clock time give a natural donor pool, and the standardised surprise is the quantity market participants already track against consensus. Three lessons transfer. First, the volatility effect is far better identified than the directional one, which supports managing event risk through position size and hedges rather than

through forecasts of the print. Second, pre-release drift is a warning: a simple before/after comparison on an event day would have mistaken positioning for reaction. Third, magnitudes change over time (2022 versus later), so any estimate should be refreshed rather than used as a constant.

Reproducibility. All figures and numbers are produced by notebooks/01_btc_us_macro_causal.ipynb and report/build_report.py (fixed seeds, requirements.txt); the notebook runs locally or on Google Colab (first cell clones the repository and downloads the Binance data). Repository: github.com/lschwob/causal-inference-proj.

References. *Abadie, A., Diamond, A., Hainmueller, J. (2010), Synthetic control methods for comparative case studies, JASA. Bauer, M., Swanson, E. (2023), A reassessment of monetary policy surprises and high-frequency identification, NBER Macroeconomics Annual. Doudchenko, N., Imbens, G. (2016), Balancing, regression, difference-in-differences and synthetic control methods, NBER WP 22791. Rambachan, A., Roth, J. (2023), A more credible approach to parallel trends, REStud. Course notes, sessions 1–4 (MSA-DAT09-04).*

Appendix A. One-page recommendation

Decision. Should a trading desk or an investor treat scheduled US macro releases (CPI, NFP, FOMC) as a risk trigger on Bitcoin exposure, and can it draw a directional signal from them?

Answer. Yes for risk. For direction, the sign is stable but the magnitude is uncertain: useful for managing risk, not yet for a trading strategy.

1. Causal findings

- **Volatility.** In the 30 minutes after a release, realised volatility is ×2.4 that of the same minutes on comparable days without a release (DiD over 151 releases, t = 17; CPI ×2.7, NFP ×2.5, FOMC ×1.9). 41–44% of releases are individually significant against in-space placebos (5% expected by chance). The effect disappears when the clock is shifted by ±6 or 12 hours.
- **Direction.** Measured against the synthetic day, the reaction follows the surprise: −65 bps per standard deviation at 5 minutes and −80 at 30 minutes (128 releases, permutation $p < 0.001$). A number that is more hawkish than expected (higher payrolls, higher core CPI, a tighter Fed) lowers Bitcoin. The sign is negative in all 20 sub-samples; the slope is −98 in 2022 and −32 after.
- **Before the release.** Bitcoin rises about 23 bps in the two hours before, mostly in the last hour. Parallel trends are rejected ($p = 0.002$), so the equal-weight DiD is not used for signed effects.

2. What we can and cannot claim

- **Can claim:** releases cause a large short-run increase in Bitcoin volatility; the short-run price reaction goes in the direction “good macro news is bad news for Bitcoin”.
- **Cannot claim:** a precise magnitude (surprises are not comparable across types, CPI is core only, FOMC has 16 surprises); that the effect is stable (it fell markedly after 2022); that the pre-release drift is caused by anticipation or can be traded.

3. Operational implications

- Reduce size or leverage on Bitcoin before 08:30 New York on CPI and NFP days, and before 14:00 on FOMC days.
- Widen stops and risk limits for the first 30 minutes; avoid executing large orders in that window.
- Read a print above consensus as a reason to hedge a long position rather than to open a short. No systematic directional strategy until it is validated out of sample, net of execution costs.

4. Follow-up

- Validate out of sample on the next two or three releases of each type.
- Add headline CPI and FOMC surprises for 2024–2026; add a control asset (ETH, gold) to test whether the pre-release drift is specific to Bitcoin.
- Month-block bootstrap for standard errors; honest-DiD bounds as a sensitivity analysis.

Main risks to this conclusion: unobserved concurrent news, regime change since the spot ETFs, and imperfect public consensus data.